



## Pool boiling CHF enhancement by micro/nanoscale modification of zircaloy-4 surface

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### ARTICLE INFO

#### Article history:

Received 13 November 2009

Received in revised form 21 June 2010

Accepted 6 July 2010

### ABSTRACT

Consideration of the critical heat flux (CHF) requires difficult compromises between economy and safety in many types of thermal systems, including nuclear power plants. Much research has been directed towards enhancing the CHF, and many recent studies have revealed that the significant CHF enhancement in nanofluids is due to surface deposition of nanoparticles. The surface deposition of nanoparticles influenced various surface characteristics. This fact indicated that the surface wettability is a key parameter for CHF enhancement and so is the surface morphology. In this study, surface wettability of zircaloy-4 used as cladding material of fuel rods in nuclear power plants was modified using surface treatment technique (i.e. anodization). Pool boiling experiments of distilled water on the prepared surfaces was conducted at atmospheric and saturated conditions to examine effects of the surface modification on CHF. The experimental results showed that CHF of zircaloy-4 can be significantly enhanced by the improvement in surface wettability using the surface modification, but only the wettability effect cannot explain the CHF increase on the treated zircaloy-4 surfaces completely. It was found that below a critical value of contact angle ( $10^\circ$ ), micro/nanostructures created by the surface treatment increased spreadability of liquid on the surface, which could lead to further increase in CHF even beyond the prediction caused only by the wettability improvement. These micro/nanostructures with multiscale on heated surface induced more significant CHF enhancement than it based on the wettability effect, due to liquid spreadability.

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## 1. Introduction

The critical heat flux (CHF) is the maximum heat flux at which nucleate boiling heat transfer sustains high cooling efficiency. After a surface heat flux reaches the CHF, it becomes coated with a vapor film that interferes with the contact between the surface and the ambient liquid, decreasing the heat transfer efficiency. The system temperature rises, and a system failure occurs if the temperature exceeds the limits of the materials in the system. For this reason, every system incorporates a safety margin by operating at a heat flux much lower than the CHF, even though this reduces the system efficiency. This compromise between safety and efficiency is an important issue in thermal systems such as nuclear power plants.

### 1.1. Increase of CHF in nanofluid

You et al. (2003) found that adding tiny amounts (less than 0.001% by volume) of alumina nanoparticles to a conventional cooling liquid significantly increased the CHF by up to 200%. However, such a large CHF enhancement in nanofluids cannot be explained using traditional CHF theories such as hydrodynamic instability models (Kutateladze, 1950; Zuber, 1959). Bang and Chang (2005) reported that boiling in a nanofluid with a flat plate heater could change the surface roughness and wettability properties due to the effect of surface modifications produced by nanoparticle deposition on the heating surface during boiling. Kim et al. (2005,2006) carried out pool boiling CHF experiments with pure water on a heater fouled by nanoparticles during pre-boiling in a nanofluid. They found that the CHF of pure water on the nanoparticles fouled surface was equal to (or sometimes higher than) the value measured in the nanofluid boiling. The results thus revealed that the CHF enhancement in nanofluids is caused by the surface deposition of nanoparticles during nanofluid boiling and the resulting changes in surface characteristics. It was also suggested that the improvement in liquid-solid interfacial parameter such as surface

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**Nomenclature**

|            |                                                                                   |
|------------|-----------------------------------------------------------------------------------|
| $q''$      | heat flux, W/m <sup>2</sup>                                                       |
| $d$        | depth of the temperature measurement point from the boiling surface, m            |
| $g$        | acceleration of gravity, m/s <sup>2</sup>                                         |
| $h_{lg}$   | latent heat of evaporation, J/Kg                                                  |
| $k$        | thermal conductivity, W/m/K                                                       |
| $U$        | uncertainty                                                                       |
| $\Delta T$ | temperature difference, °C                                                        |
| $R_a$      | Roughness, m                                                                      |
| $\Delta d$ | distance between temperature measurement points in the conduction copper block, m |

*Greek letters*

|          |                             |
|----------|-----------------------------|
| $\theta$ | contact angle, deg or rad   |
| $\rho$   | density, kg/m <sup>3</sup>  |
| $\beta$  | contact angle, deg or rad   |
| $\phi$   | angle of rotate, deg or rad |
| $\sigma$ | surface tension, N/m        |

*Subscripts*

|        |                    |
|--------|--------------------|
| CHF    | critical heat flux |
| $l$    | liquid             |
| sat    | saturation         |
| $g$    | vapor              |
| sample | zircaloy-4 sample  |
| wall   | heater surface     |

wettability could be a key factor in interpreting the physical mechanism of the nanofluid CHF increase. Kim et al. (2007) reported that nanoparticle-fouled surfaces had significantly increased wettability measured by a reduction in the static contact angle. Based on a review of the prevalent CHF theories, they suggested that

the improved wettability caused by the nanoparticle layer could predict CHF enhancement. Liu and Liao (2008) and Coursey and Kim (2008) performed pool boiling experiments with water-based and alcohol-based nanofluids on a plain heated surface. Both studies used static contact angle measurements to confirm that the nanoparticle layer formed on the heater surface significantly improved the wettability. Despite this volume of significant results related to nanofluids, there has been very little practical application to real thermal systems because of the difficulty in maintaining a stable supply of nanofluids.

### 1.2. Better surfaces for real applications

A large number of investigations for the argumentation of boiling heat transfer and CHF using surface modification have taken place since the 1950s. A summary of some examples is given in Table 1. Recently Kim et al. (2009) engineered the silicon–ZnO surfaces using micro fabrication in order to imitate the micro/nanostructures of the nanoparticles deposition layer that resulted in the significant CHF enhancement of the nanofluid boiling. The various surfaces with flat, micro, nano, micro/nano-patterned ZnO structures showed quite significant but diverse CHF increase results, which could be correlated using a combination of the surface roughness and surface wettability. Besides the wettability effect of CHF enhancement, the liquid spreadability of the micro/nano-patterned surface resulted in the highest CHF increase. This result was quite significant, although a ZnO structure is not very robust.

To simulate a more practical application, zircaloy-4 was used in this study as the test material for pool boiling CHF experiments. This is the same material used in nuclear fuel rod cladding. The aim was to enhance the CHF on the surface of zircaloy-4, rather than the surface of normal metals such as copper and stainless steel. Anodic oxidation (anodization) was used change the morphology and wettability of zircaloy-4. These modified zircaloy-4 surfaces were then tested in pool boiling experiments (Fig. 1).

**Table 1**

Summary of modified surface on CHF enhancement reported in the literature.

| Reference                      | Heater          | Working fluid and condition | Modification method                                               | Surface characteristics                   |
|--------------------------------|-----------------|-----------------------------|-------------------------------------------------------------------|-------------------------------------------|
| Costello and Frea (1965)       | SUS cylinder    | Distilled, tap water, 1 bar | Capillary wicking material inserting and deposition               | Wettability improve                       |
| Hahne and Diesselhorst (1978)  | Pt, NiCr tube   | Deionized water, 1 bar      | Oxidized surface by repeated experiment                           | Wettability improve                       |
| Messina and Park (1981)        | Cu plate        | Freon 113, 1 bar            | Etching with pit arrays, sanding, mirror by polishing             | Surface micro geometry                    |
| Marto and Lepere (1982)        | Cu tube         | FC72, Freon 113, 1 bar      | Copper enhanced surface with cavities                             | Cavity size control                       |
| Chowdhury and Winterton (1985) | Al, Cu cylinder | Water, methanol, 1 bar      | Roughness, surface energy (anodization)                           | Wettability, Ra improve,                  |
| Liaw and Dhir (1986)           | Flat copper     | Water, Freon 113, 1 bar     | Oxidation and fluoro-silicone coating                             | Wettability improve                       |
| Anderson and Mudawar (1989)    | Cooper          | FC-72, 1 bar                | Roughness, artificial cavity (fins, studs, grooves)               | Roughness improve                         |
| Golobic and Ferjancic (2000)   | Ti, Fe ribbon   | FC-72, 1 bar                | Roughness (sandpaper), etching on low roughness, particle coating | Roughness improve                         |
| Fong et al. (1999)             | Zircaloy-2 tube | Water, 1 bar                | Created micro size cavities, contact angle (glass bead-pined)     | Wettability, Ra improve                   |
| Liter and Kaviany (2001)       | Not reported    | Not reported                | Modulated porous layer coating                                    | Wettability improve                       |
| Ferjancic and Golobic (2002)   | Fe ribbon       | Water, FC-72, 1 bar         | Roughness change, etched surface (sanding)                        | Ra improve                                |
| Kim et al. (2002)              | Pt Wire         | FC-72, 1 bar                | Diamond particles—micro-porous layer coating                      | N/C                                       |
| Honda et al. (2002)            | Si chip         | FC-72, 1 bar                | Dry etching (micro-fins, sub-micron roughness)                    | Micro structure                           |
| Fong et al. (2003)             | Zr-2 tube       | Water, 1 bar                | Oxidation                                                         | Wettability, Ra improve                   |
| Takata et al. (2003)           | Copper plate    | Water, 1 bar                | TiO <sub>2</sub> coating and UV irradiation                       | Wettability improve                       |
| Ujereh et al. (2007)           | Si and Cu       | Water, 1 bar                | Aged with UV irradiation                                          | Well aged improve                         |
| Kim et al. (2009)              | Si (Ti film)    | Water, 1 bar                | Micro and nano structure (ZnO on silicon)                         | Wettability improve, micro/nano structure |

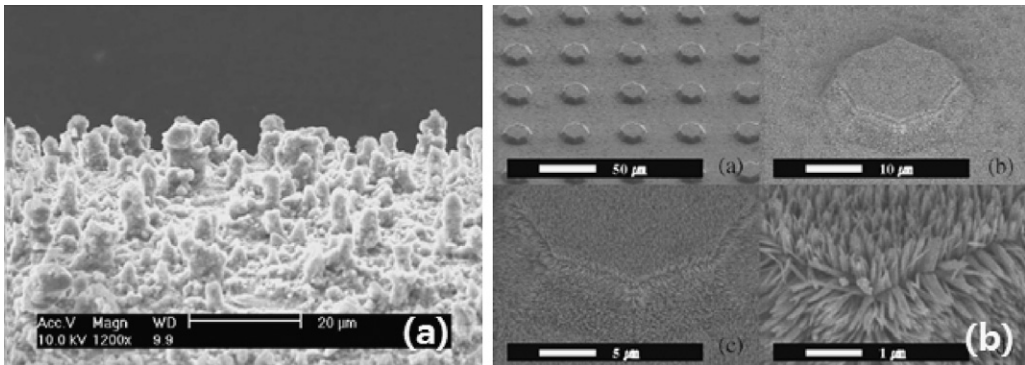


Fig. 1. (a) Nanoparticle-coated NiCr wire after pool boiling CHF experiment with nanofluids (Kim et al., 2005) and (b) surface modification on silicon wafer (Kim et al., 2009).

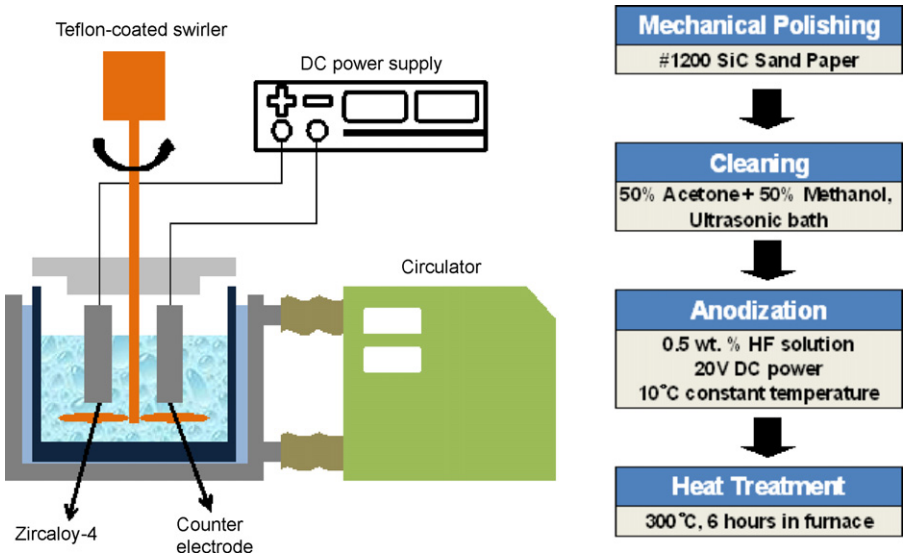


Fig. 2. (a) Anodization experimental setup and (b) the treatment process of zircaloy-4.

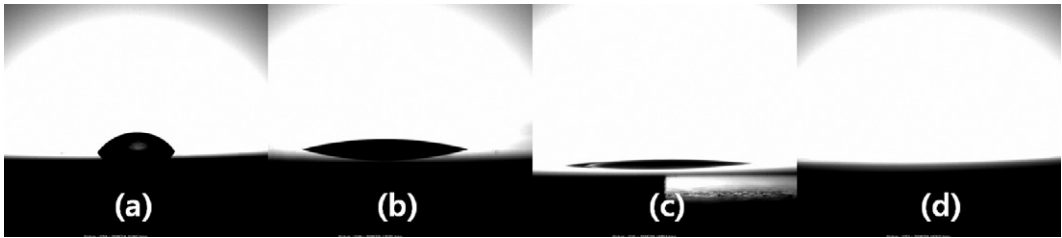


Fig. 3. Contact angle change on the treated zircaloy-4 surface (a) 49.3°, (b) 29.4°, (c) 8.4° and (d) 0°.

Table 2  
The surface characteristics of the typical test samples and CHF data.

| Anodization time (s) | Contact angle (°) | Average roughness (μm) | Spreading | ONB (°C) | CHF (kW/m <sup>2</sup> ) |
|----------------------|-------------------|------------------------|-----------|----------|--------------------------|
| 600                  | 0                 | 0.32                   | Large     | 121      | 1924                     |
| 600                  | 0                 | 0.31                   | Large     | 123      | 1770                     |
| 300                  | 2.1               | 0.07                   | Small     | 134      | 1406                     |
| 180                  | 3.2               | 0.05                   | Small     | 130      | 1669                     |
| 60                   | 9.1               | 0.15                   | Small     | 132      | 1352                     |
| 45                   | 14.6              | 0.11                   | None      | 121      | 1167                     |
| 30                   | 15.6              | 0.13                   | None      | 121      | 1148                     |
| 15                   | 25.7              | 0.16                   | None      | 123      | 1240                     |
| 15                   | 29.4              | 0.15                   | None      | 122      | 1272                     |
| N/C                  | 49.3 (bare)       | 0.15                   | None      | 121      | 1004                     |

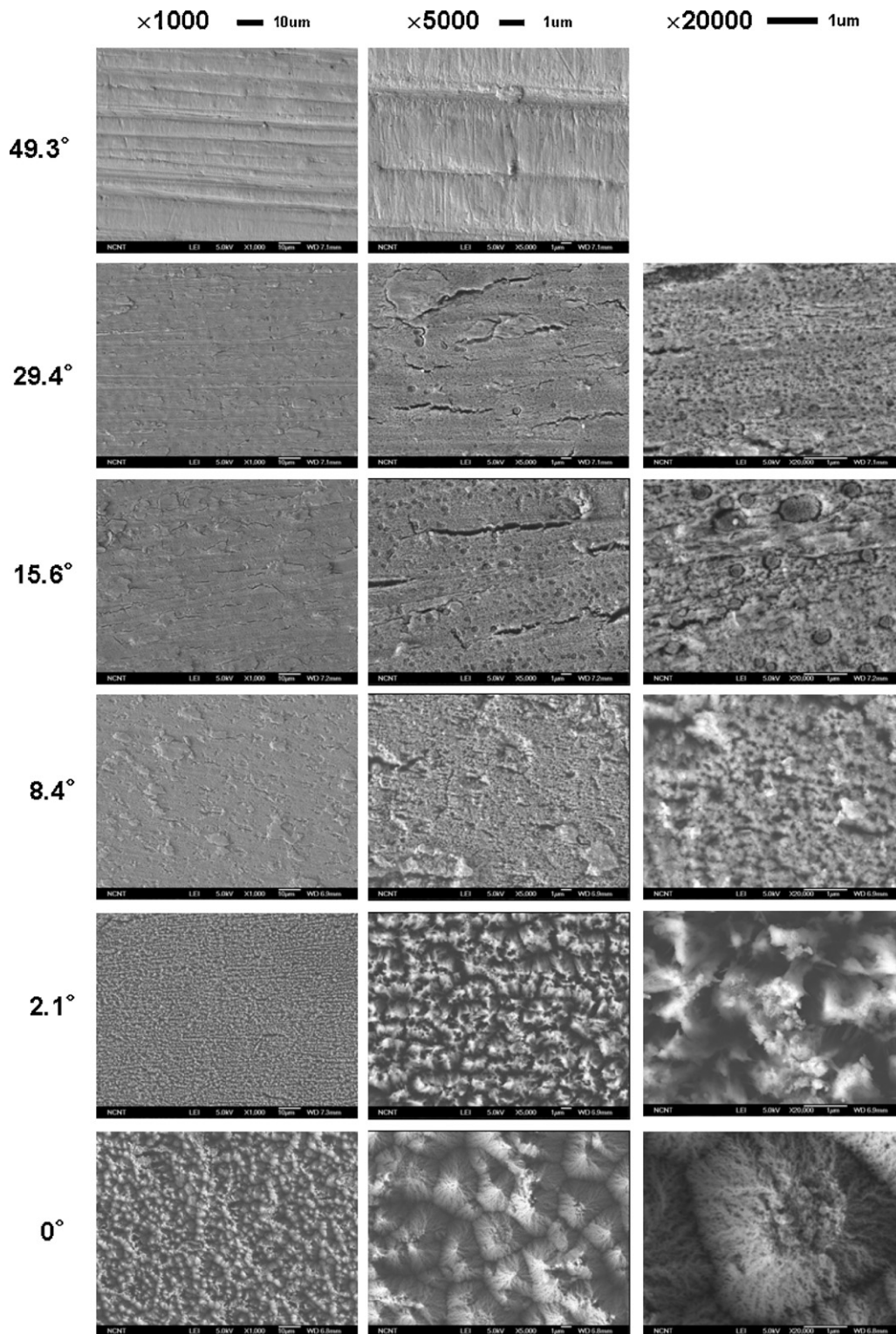


Fig. 4. SEM image of the test samples.

## 2. Experimental preparation

### 2.1. Preparation of test surfaces

Anodic oxidation is an electrochemical fabrication technique in which an external applied electric potential is used to enhance the chemical reaction at the interface between an electrolyte and the target material. The reaction rate can be very easily controlled

by adjusting the applied voltage; this is as sensitive as a 108-fold increase of the reaction rate per volt of increased potential (Soriaga, 1988). Varying the other parameters such as current density, type of electrolyte, and acid strength also affect the reaction and result in a different end product. The final product of anodic oxidation is a generally a thick oxide layer. However in certain materials and under certain conditions, unique ordered nanostructures occur instead of randomly formed oxide structures. Anodic aluminum



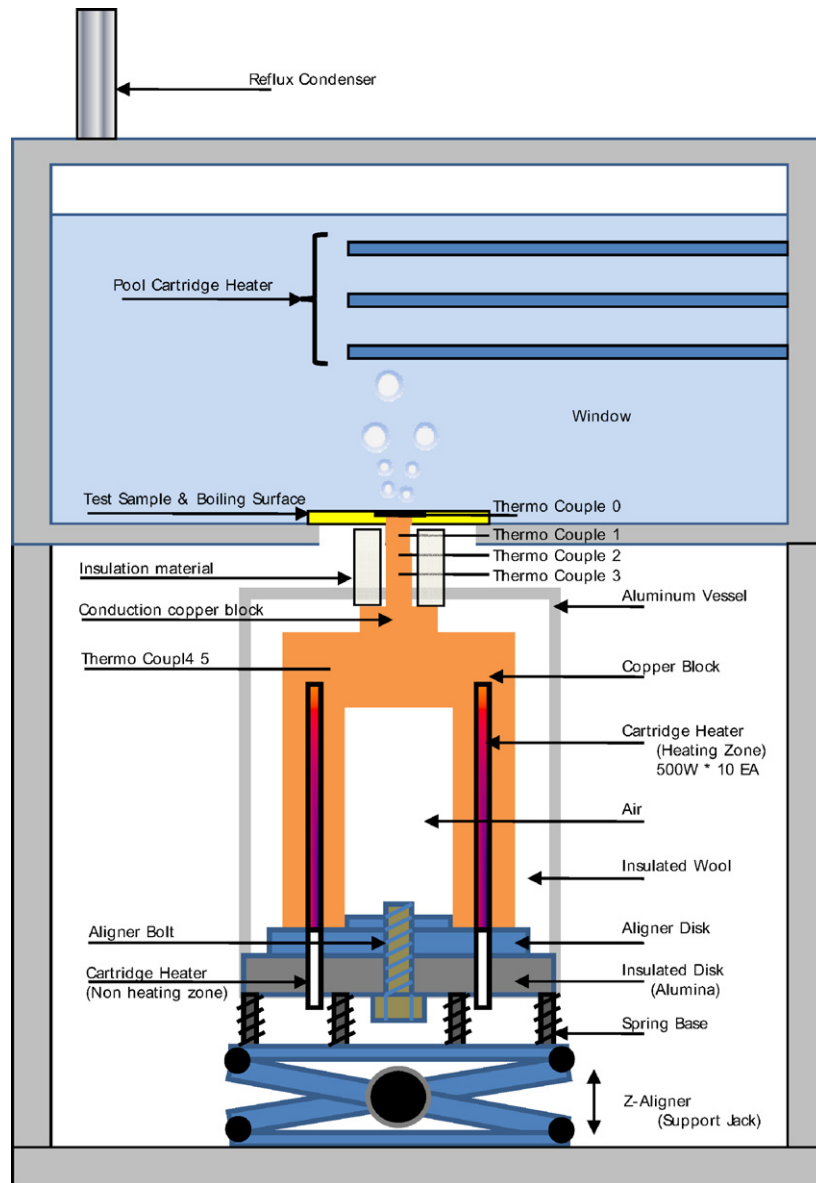


Fig. 5. The schematic diagram of pool boiling CHF facility.

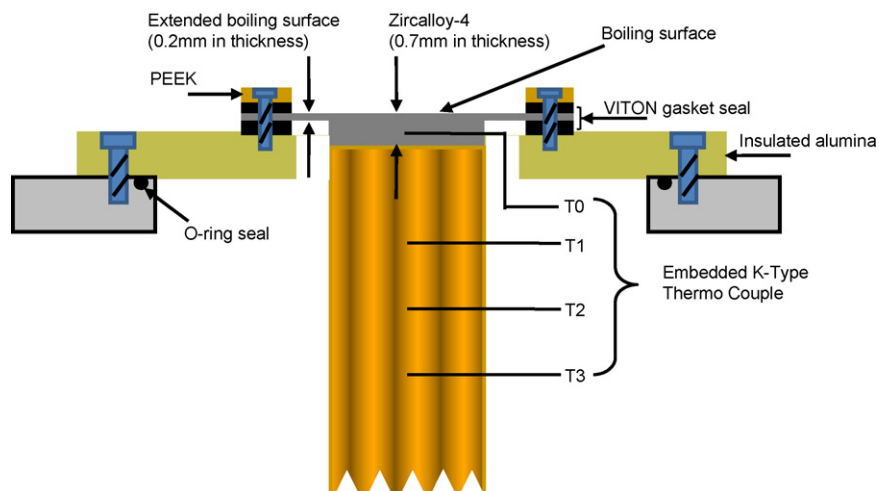


Fig. 6. The detailed sketch of the test section.

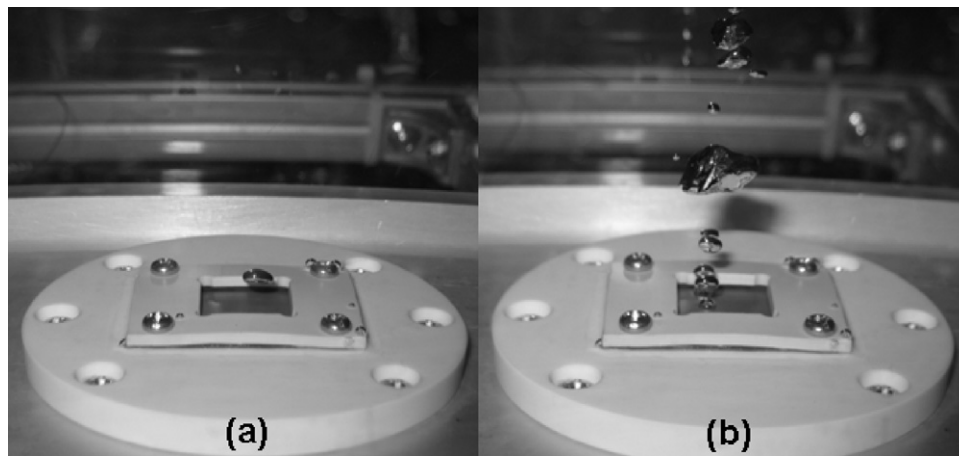


Fig. 7. Boiling pictures near the ONB (a) bare zircaloy-4 (wallsuperheat: 22 °C) and (b) treated zircaloy-4 (wallsuperheat: 28 °C).

oxide (AAO) (Ono et al., 2005) and titania nanotubes (Gong et al., 2001) are two well-known examples. Because of their nanoporous surface morphologies and the chemical nature of the metal oxide, these examples are very hydrophilic. Anodic oxidation of zircaloy has already been attempted and the results have been observed by microscopy (Ploc and Miller, 1977). However, no ordered nanostructures have been found and few studies have attempted to measure the resulting wetting properties. Ever since titania nanotubes were first reported in the early 2000s, we have followed the progress of related research very closely because both titanium and zirconium are in the same chemical group in the periodic table, and similar forms can be expected for zirconium alloy. Zirconia nanotubes were indeed subsequently reported by Lee and Smyrl (2008). After in-depth study of the preceding results, we finally succeeded in fabricating zircaloy nanotubes. We then conducted wettability studies on these nanotubes as well as the associated micro/nanoscale mountain-like structures that were subsequently discovered.

Rectangular zircaloy-4 plates (20 mm × 25 mm × 0.7 mm) were used as test samples. They were mechanically polished with #1200 silicon carbide abrasive to remove impurities and produce a uniform surface. The polished samples were cleaned with a 1:1 mixture of acetone and methanol in an ultrasonic bath. After rinsing in deionized water, the samples were completely dried (Salot et al., 1996). Fig. 2 shows the setup used for the anodization process where 0.5 wt.% hydrofluoric acid solution was used as the electrolyte. The sample surfaces were anodized by applying a constant electric potential (20 V) between the anode and the cathode using a DC power supply (N5771A; Agilent). The solution temperature was fixed at 10 °C to maintain a constant reaction speed. The exposure time to the anodization process was varied from 0 to 600 s to obtain various zirconium oxide surfaces with different surface morphologies. After the anodization process, the treated samples were heated for 6 h at 300 °C in an electric muffle furnace to eliminate any fluoride residue (Lee and Smyrl, 2008).

## 2.2. Characterization of test surfaces

In this study, the wettability of zircaloy-4 was controlled by the exposure time in the solution (anodization time). As the exposure time in the solution was longer, the contact angle was lower. Fig. 3 shows the contact angles of deionized water on the prepared sample surfaces. The contact angle measurement was done with 1  $\mu$ L of the deionized water. All of the pictures were taken 5 s after dispensing. Measurements were done on at least five points for every sample. Highest and lowest 20% data were trimmed out before tak-

ing average. The uncertainty of contact angle measurement was  $\pm 2^\circ$ .

As examples in the previous studies, anodization of zircaloy-4 can create some meaningful micro/nanoscale structures in some conditions. To find the conditions, we tested a number of different anodization conditions with zircaloy-4 and examined the resulting surface structures using scanning electron microscope (SEM) as seen in Fig. 4. The exposure time in the solution (anodization time) was longer, the contact angle was lower and some structures on zircaloy-4 were developed. The longer duration of anodization time in the solution, the lower contact angle was shown and some nanoscale structures on zircaloy-4 were developed. At the bare surface (with the contact angle of 49.3°) and the treated surface (with the contact angle from 15.6° to 29.4°), no meaningful surface structures, but only oxide layer were found. At the treated surface (with the contact angle less than 10°), the meaningful microstructure was still not found. However, at the treated surface with the contact angle near 0, the meaningful microstructures like the uniform valley were found on the SEM image.

After the treatment of zircaloy-4, the surface roughness of all zircaloy-4 samples was measured by the surface profiler (Alpha-Step IQ; KLA Tencor). The average roughness data of the bare zircaloy-4 was different than that of the treated zircaloy-4 (Table 2). Measurements were performed for at least five points on every sample. The peak average roughness ( $R_a$ ) of the treated zircaloy-4 surface was less than 0.35  $\mu$ m and the peak  $R_a$  of the bare zircaloy-4 surface was less than 0.16  $\mu$ m. These sub-micron-sized  $R_a$  changes of the surface could not account for the CHF enhancement. According to Hahne and Grigull (1977), the surface roughness affects the vapor bubble growth due to the distribution and activation of nucleation sites, but the effects of changing density of active nucleation sites are relatively weak near the peak heat flux. This was mainly confirmed by the fact that the roughness shows no effect on the CHF values (Tong, 1968).

## 2.3. Experimental setup

Fig. 5 shows a schematic diagram of the pool boiling experimental facility. It consists of a boiling pool part, test sample, and heating section. The boiling pool part has a test pool, immersion heaters, visualization windows and reflux condensers. The aluminum test pool is a rectangular vessel with a height of 600 mm. Three immersion heaters with the maximum heating power of 3 kW are embedded in the pool to preheat working fluid. A 10 mm-thickness polycarbonate window with a maximum allowable temperature of 140 °C is placed in the front and back wall of the

pool to visualize the boiling phenomenon on the test heater. A reflux condenser is located at the top of the pool to condense phase-changed vapor into liquid. Three T-type thermocouples are immersed into the pool to measure the average bulk temperature. During the experiment, the bulk temperature is maintained at 100 °C using a feedback control of the thermocouples and cartridge pool heaters.

The test sample is at the bottom of the pool, as shown in Fig. 5. For boiling experiments, the sample is heated by conduction from the heating section, which consists of conductive copper blocks and ten cartridge heaters with an individual maximum heating power of 500 W. The cartridge heaters with total capacity of 5 kW heat the copper block, and the generated thermal energy is transferred to the test sample via conduction. All parts of the heating section are sustained by uniformly spaced mechanical springs on the supporting jack, and thus small and uniform thermal contact resistance is guaranteed at the interface between the copper block and test sample. When the CHF phenomenon occurs on the heating surface, the heated copper block is immediately lowered using the supporting jack, so the direct contact to the heating surface is removed. This mechanism helps to protect the test sample from thermal damage caused by the tremendous temperature increase right after the CHF.

Fig. 6 shows the detailed schematics of the assembly of the rectangular zircaloy-4 test sample and the conductive copper block. To prevent a leakage of working fluid into the heating section, a VITON gasket with the maximum temperature of 210 °C is inserted between the test sample and the zirconia block with low thermal conductivity ( $\sim 3 \text{ W/m/K}$ ). The micro-sized gaps between the boiling surface and the side insulation structures would be served as the preferred cavities of bubble nucleation, and then premature bubble nucleation from these undesired cavities could severely distort boiling heat transfer characteristics, resulting in poor repeatability of experiments. Thus, the test sample is post-machined to have the actual boiling surface (10 mm  $\times$  10 mm) only at the center of the test sample and the thin fin (0.2 mm in thickness) in the boundary, so that onset of the nucleate boiling starts on the boiling surface of interest as shown in Fig. 7. Consequently, the clear bubble nucleation phenomenon results in very good repeatability in pool boiling curves for pure water, which will be explained in Section 3.

#### 2.4. Data gathering and measurement uncertainty

To measure the surface temperature and heat flux, a 0.25-mm-diameter K thermocouple was inserted horizontally at the center of the test sample 0.3 mm below the boiling surface, and three 0.5-mm-diameter K thermocouples were embedded at intervals of 7 mm along the center axial line of the conductive 99.999% pure copper block (RNDKOREA). Because the thermal conductivity of the copper (about 400 W/m/K) was three orders of magnitude greater than that of the insulation material (about 0.5 W/m/K), the heat loss through the insulation was assumed to be less than 0.01%; this was confirmed by a numerical simulation using the commercial software, FLUENT 6.0. We assumed that the heat loss through the insulation was negligible in the conduction block surrounded by the thermal insulator since the thermal conductivity of the copper was three orders of magnitude higher than that of the insulation material. Note that the linear relationship among the three temperature measurements in the cylindrical copper block was examined with a numerical simulation for the steady-state heat conduction in the actual heater geometry using the commercial software, FLUENT 6.0 (Ahn et al., 2008), and the numerical simulation results confirmed that, Eq. (1) is proper to estimate the heat flux.

$$q'' = k \times \frac{\Delta T}{\Delta x} = k \times \frac{T_2 - T_1}{\Delta x}, \quad (1)$$

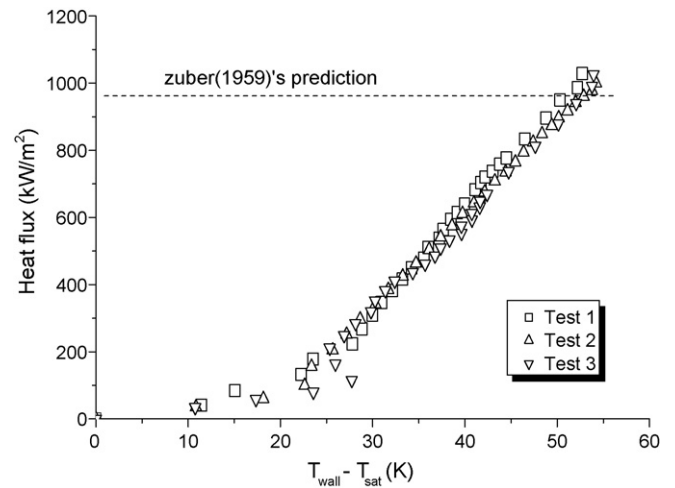


Fig. 8. Boiling curve on the bare zircaloy-4.

where  $T_1$  and  $T_2$  are the temperatures recorded in the conductive copper block. The surface temperature or wall superheat was calculated by extrapolating the temperature  $T_0$  measured inside the test sample using the heat flux obtained from Eq. (1):

$$\Delta T_{\text{wall}} = T_{\text{wall}} - T_{\text{sat}} = \left( T_0 - \frac{q'' d}{k} \right) - T_{\text{sat}}. \quad (2)$$

The uncertainties in the heat flux and wall superheat measurements were (Kline and McClintock, 1953):

$$\frac{U_{q''}}{q''} = \sqrt{\left( \frac{U_{T_2 - T_1}}{T_2 - T_1} \right)^2 + \left( \frac{U_{\Delta x}}{\Delta x} \right)^2} \quad (3)$$

and

$$\frac{U_{\Delta T_{\text{wall}}}}{\Delta T_{\text{wall}}} = \sqrt{\left( \frac{U_{T_0 - T_{\text{sat}}}}{T_0 - T_{\text{sat}}} \right)^2 + \left( \frac{U_{q''}}{q''} \right)^2 + \left( \frac{U_d}{d} \right)^2}. \quad (4)$$

where the parameters  $U_{q''}$ ,  $U_{T_2 - T_1}$ , and  $U_{\Delta x}$  are the uncertainties of the heat flux  $q''$ ,  $T_2 - T_1$ , and  $\Delta x$ , respectively. All the thermocouples were calibrated using a reference resistance temperature detector sensor with a measurement accuracy of  $\pm 0.2 \text{ K}$  certified by the Korea Testing Laboratory. The overall maximum uncertainties in the heat flux and wall superheat measurements were 10.2% and 11.2% at  $q'' = 100 \text{ kW/m}^2$ , and 2.3% and 3.2% at  $q'' = 900 \text{ kW/m}^2$ .

### 3. Experimental results

Fig. 8 shows the pool boiling heat transfer results for deionized water on bare zircaloy-4 surfaces. The nucleate boiling heat transfer rate and CHF point agreed closely within  $\pm 5\%$  for the three tests. Moreover, the CHF values for the bare surface were quite close to the CHF correlation predicted by Zuber (1959),

$$q''_{\text{CHF-Zuber}} = 0.131 h_{\text{lg}} \rho_{\text{lg}}^{0.5} [\sigma g (\rho_l - \rho_g)]^{0.25}. \quad (5)$$

This indicates good control of the experimental conditions, such as the degassing of dissolved gas, saturation temperature of the working fluid, and test sample surface roughness and wettability. It also indicates that the nucleation of the premature bubbles in the undesired cavities of the heater-insulator interface was successfully inhibited (Fig. 7).

Fig. 9 shows pool boiling curves on the bare and treated zircaloy-4 surfaces using deionized water for contact angles in the range of 49.3° (bare) to 0°. The onset of nucleate boiling and the boiling heat transfer performance of all samples are indicated. All cases exhibited almost identical nucleate boiling heat transfer characteristics.

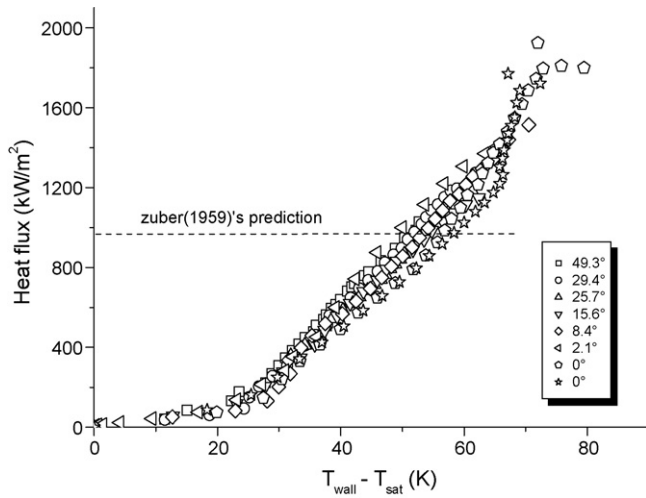


Fig. 9. Boiling curve on the treated zircaloy-4.

The onset of nucleate boiling occurred at a similar wall superheat of approximately 21 °C. In addition, CHF of treated zircaloy-4 over the whole range of contact angle (i.e. 0–40°) increased with 20–90% compared with bare zircaloy-4. As Table 2 shows, the average roughness change of the treated zircaloy-4 was a function of the anodization time. However, the difference in roughness was very small, on the sub-micron level, between the bare zircaloy-4 and the treated zircaloy-4.

#### 4. Data interpretation

##### 4.1. Changes in contact angle

The contact angle of zircaloy-4 treated by anodization was controlled by the processing time, as shown in Table 2. These changes in the surface contact angle were related to changes in the surface wettability, which can affect the CHF enhancement. There is general consensus about the effect of wettability on the CHF.

Kandlikar (2001) analyzed the mechanism of CHF phenomena using a balance of the recoil force and the surface tension on the heating surface. The recoil force was generated by fast evaporation into liquid–vapor interface. The surface tension on the heating surface was the liquid property which included the effect of the surface wettability (contact angle). He also suggested the following correlation model:

$$q''_{CHF} = h_{fg} \rho_g^{1/2} \left( \frac{1 + \cos \beta}{16} \right) \left[ \frac{2}{\pi} + \frac{\pi}{4} (1 + \cos \beta) \cos \phi \right]^{1/2} \times [\sigma g (\rho_l - \rho_g)]^{1/4} \quad (6)$$

Fig. 10 shows the CHF enhancement achieved in this study compared with the model prediction of Kandlikar and the experimental data of Hahne and Diesselhorst (1978). The CHF enhancement achieved in this study, which was dependent on the contact angle, displayed a similar tendency to the theoretical model and to previous experimental data. Fig. 11 shows the CHF enhancement ratio (%) versus the bare case which was non-treated zircaloy-4 with the contact angle of 49.3°. In addition, Kandlikar's prediction was used to present the CHF enhancement ratio (%) as the reference data with the contact angle of 49.3°. The treated zircaloy-4 with contact angles from 10° to 49.3° enhanced the CHF in a manner quite close to Kandlikar's correlation (Figs. 10 and 11). On the other hand, for contact angles less than about 10°, the CHF enhancement related to the contact angle change was greater than that predicted by Kandlikar (Figs. 10 and 11).

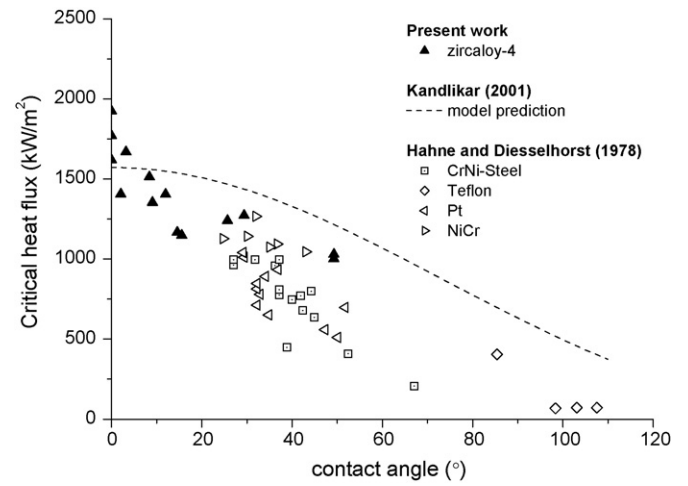
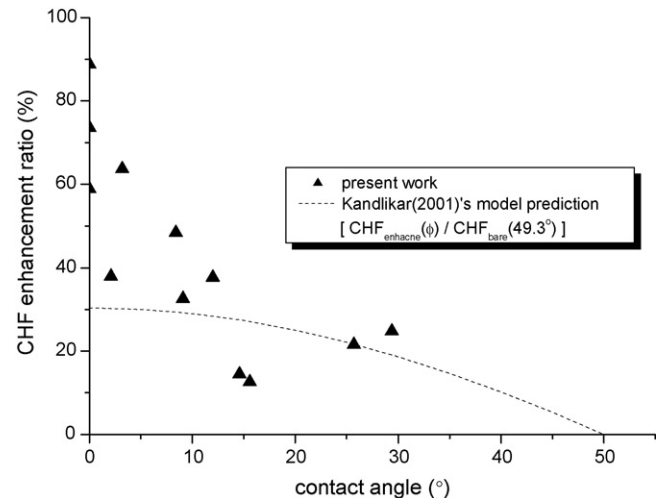


Fig. 10. CHF enhancement on the treated zircaloy-4.

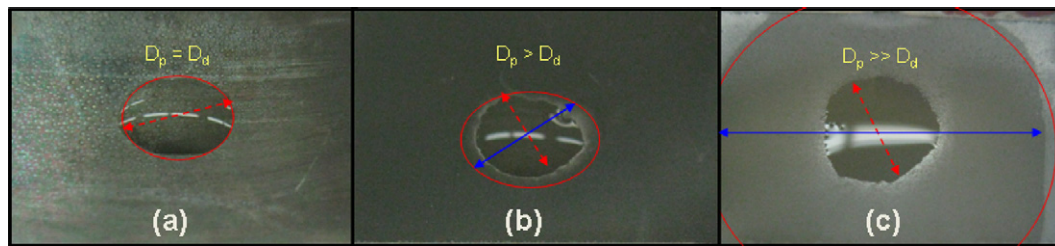
##### 4.2. Surface capillarity

In the high heat flux region near the CHF, the ability to rewet the dry region generated by the numerous vapor patches that occur during the phase change process is very important to the growth of the dry region through the CHF (Theofanous et al., 2002). Furthermore, the vapors generated mushrooms with coalescence vapor columns near the CHF (Sadasivan et al., 1992; Haramura and Katto, 1983). Rewetting the dry region in the large mushrooms with the liquid at the bulk temperature is difficult, just as cooling the heated surface is also difficult. As these mushrooms grow, the ability to rewet the dry region from where the mushrooms depart could delay the CHF phenomenon (Theofanous and Dinh, 2006). However, some questions still exist as to the mechanism of the liquid inflow that could wet the dry region. According to Kandlikar's CHF model, there must be almost no CHF enhancement for contact angle changes of less than 10°.

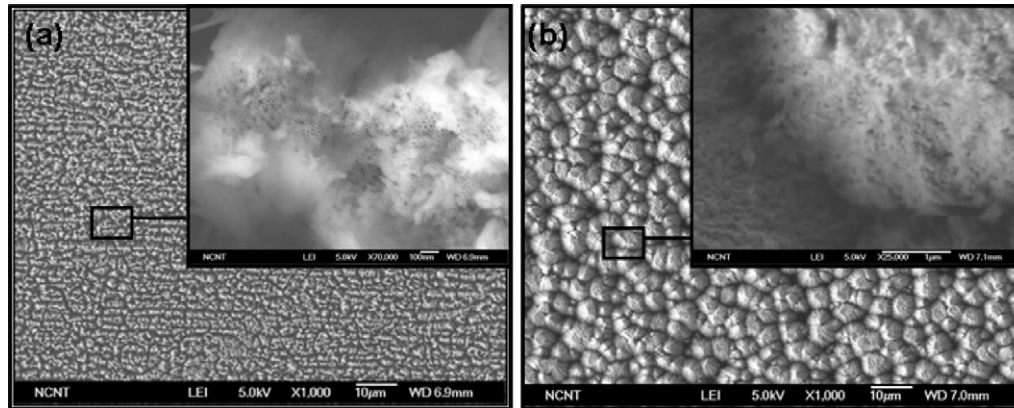
During the process of contact angle measurement, we could find out the liquid spreading phenomenon on the prepared sample surfaces like to Fig. 12. To know the difference of the liquid spreading ability, the 10 µL droplet of the deionized water dropped on the treated zircaloy-4 surface. All pictures in Fig. 13 were taken when the droplet of water just made a contact with the surface and after 5 s from the moment.  $D_d$  was the diameter of the contact line

Fig. 11. CHF enhancement ratio on the treated zircaloy-4 ( $CHF_{\text{enhance}}/CHF_{\text{bare}}$ ).





**Fig. 12.** The liquid spreading abilities of each treated zircaloy-4 ( $D_p$ : the precursor line of droplet,  $D_d$ : the contact line of droplet) (a) non-spreading, (b) small spreading and (c) large spreading.



**Fig. 13.** (a) Small spreading with contact angle of  $2.1^\circ$ , nano size tube and (b) large spreading with contact angle of  $0^\circ$ , nano size thread.

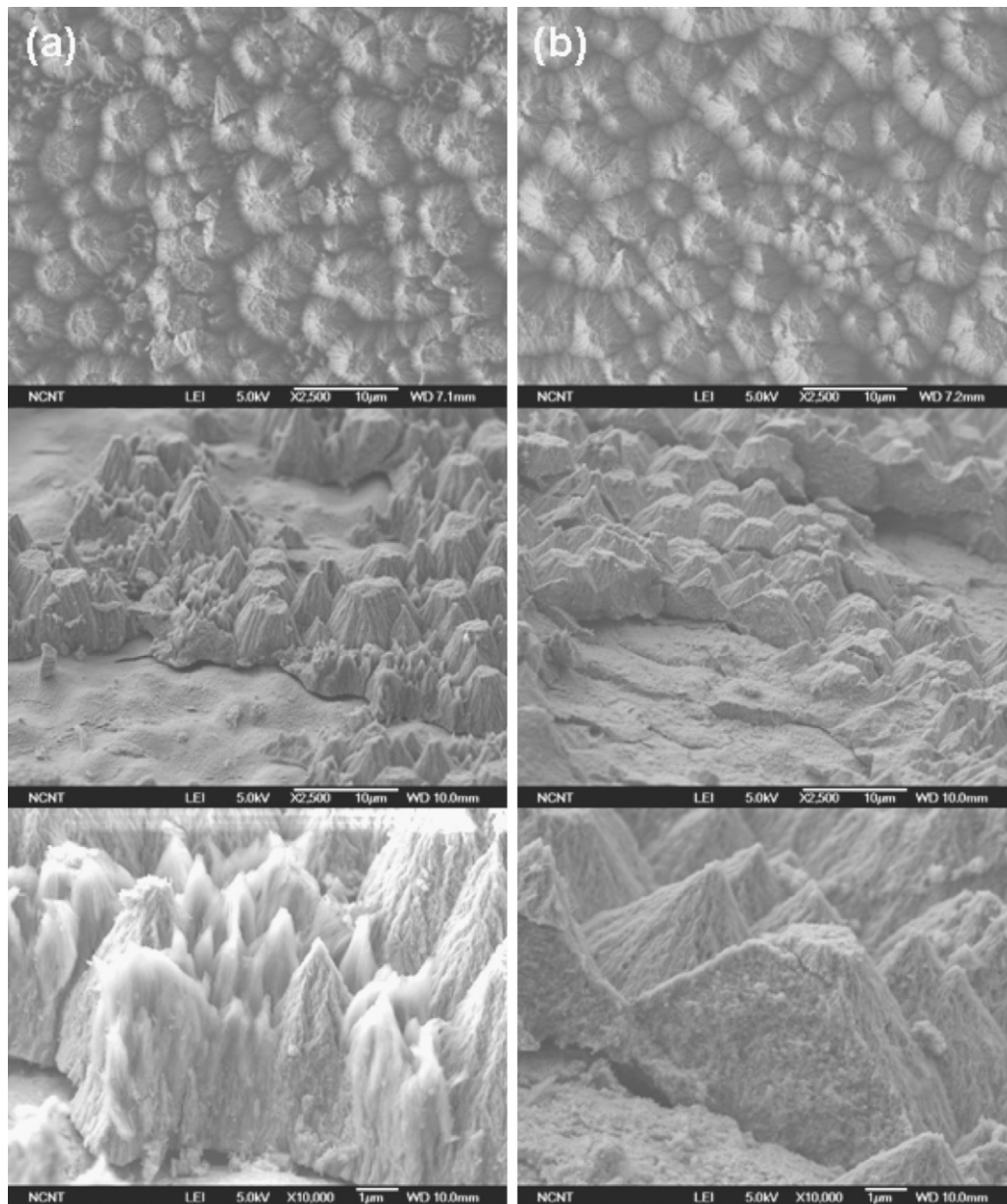
between the droplet and the surface and  $D_p$  was the diameter of the precursor line of droplet on the surface. They are not accurate and measurable values. However,  $D_d$  and  $D_p$  were used for the relative comparison of liquid spreading in Fig. 12 (none, small and large). As shown in Table 2, the anodization time was longer, the contact angle was lower. These samples with the lower contact angle (super hydrophilic) had the liquid spreading phenomenon, but the quantities of the liquid spreading were each different. These different liquid spreading abilities of the treated zircaloy-4 could influence the CHF enhancement and have the possibility to play an important role to enhance the CHF with the contact angle less than  $10^\circ$ .

Kim and Kim (2007) showed that there is an additional factor beyond just the contact angle: capillary wicking. They explained that capillary wicking induces lateral liquid spreading to rewet the dry region beneath the vapor bubbles, delaying the triggering of the CHF. Thus, the capillary wicking (or liquid spreading) capability is another surface characteristic to be studied. Liquid spreading on micro/nano multiscale structures under high heat flux causes liquid inflow on the dry spot, which delays the growth of the dry patch that triggers the CHF phenomenon. However, in this study, there was no method of measuring the capillarity of the plate heater because the heater was not thin and cylindrical (Kim and Kim, 2007). Chen et al. (2008) reported that the capillarity on the surface that induced the spreading phenomenon like capillary wicking was surrounded by a wetted zone, as shown in Fig. 12(b) and (c). Kim et al. (2009) also conjectured that the spreading and the contact angle jointly affected the wettability of the surface and each other. The samples with contact angles of less than  $10^\circ$  exhibited the spreading phenomena shown in Fig. 12(b) and (c). For a contact angle near  $0^\circ$ , severe spreading occurred, with the completely wetted droplet surrounded by the largest wet zone, as shown in Fig. 12(c). For contact angles in the range of  $0^\circ$  to  $10^\circ$ , the spreading phenomenon was such that the droplet was surrounded by a slightly wet zone, as shown in Fig. 12(b). The sample with the large spreading ability and a contact angle of  $0^\circ$  had the highest CHF value. In addition, the samples with small spreading ability and contact angles in the

range of  $0^\circ$  to  $10^\circ$  had higher CHF values than predicted by Kandlikar, as shown in Figs. 10 and 11. The experimental results showed that a large spreading ability could dramatically enhance the CHF well above the prediction of Kandlikar's model. The liquid spreading ability in the range of contact angles from  $0^\circ$  to  $10^\circ$  induced capillary wicking on the surface, and the capillary wicking influenced the fast liquid inflow to a dry spot, which in turn delayed the CHF.

#### 4.3. Effect of micro and nano surface morphology

The question remains as to what caused the spreading phenomenon on the treated zircaloy-4. Previous research has concluded that CHF enhancement in nanofluid pool boiling is due to nanosized particles fouling the heating surface to form hydrophilic micro/nanostructures. (Kim et al., 2005,2006) To estimate the wettability of the nanoparticles fouling the heating surface, the contact angle must be very low, less than  $10^\circ$ . This produces results similar to the treated zircaloy-4 reported here, which had liquid spreading ability. The CHF for nanoparticle-fouled heating surfaces is also greater than that predicted by Kandlikar's model (Kandlikar, 2001). Based on these previous research results, Kim et al. (2005,2006) reported that the significant CHF enhancement in nanofluids was due to the capillary wicking effect caused by the structures of fouled nanoparticles. This capillary wicking effect could help spread liquid on the heating surface near the CHF condition. Figs. 4 and 13 show SEM images of the treated zircaloy-4 with spreading ability at low-contact angles of less than  $10^\circ$ . (There were no microstructures worthy of notice on the treated zircaloy-4 with small spreading ability.) Nanotube-like nanostructures can be observed in Fig. 13. These nanostructures aided the liquid spreading ability of the treated zircaloy-4 to some extent and played a role in the CHF enhancement. The same surfaces also contained valley-like microstructures, as shown in Fig. 4. Chen et al. (2009) reported that microscale structures could enhance capillary wicking and capillary-driven flow. Ishino et al. (2007) reported that



**Fig. 14.** SEM image of large spreading (a) nanotube and valley micro structures (CHF: 1924 kW/m<sup>2</sup>) and (b) valley micro structures (CHF: 1770 kW/m<sup>2</sup>).

microscale structures could influence liquid capillary wicking on the surface, and the liquid spreading ability could be characterized by the diameter of the wetted zone that surrounded the liquid droplet. Kim et al. (2009) reported that nanostructures could help the liquid spreading and delay the CHF. In addition, they reported that the combination of nano- and microstructures had the best liquid spreading ability and the best enhanced CHF. The micro/nano multiscale structures that facilitated the liquid spreading on the heated surface were like a porous medium. Therefore, the combined micro- and nanoscale structures observed in Figs. 4 and 13 could play an important role in the significant CHF enhancement observed in the present study for contact angles of less than 10°, as described in Section 4.1.

The amount of CHF enhancement was different for various samples with large spreading ability. Fig. 14(a) shows nanosized tubes with small spreading ability formed between nano/micro multiscale structure valleys with large spreading ability, which have already been described; for this case, the CHF increased by about 90%. Fig. 14(b) shows only nano/micro multiscale structures of var-

ious valleys; here, the CHF increased by about 70%. These results (i.e., the CHF enhancement data and the SEM observations) indicate that it is not only the nano/micro multiscale structures that significantly influence the CHF enhancement, but also the combination of the two different nano/micro structures. Moreover, we contend that the latter could affect the CHF enhancement more significantly than the former because of the combined effects of these multiple structures.

## 5. Conclusion

Pool boiling CHF enhancement by anodization of zircaloy-4 surfaces was investigated under saturated and atmospheric conditions. Zircaloy-4 was selected to be relevant to real application because it is used as cladding material of fuel rods in nuclear power plants. The anodization of the zircaloy-4 surfaces decreased contact angle of a water droplet by creating micro/nanostructures from 49.3° to 0°. The findings obtained in the present study are as follows:

- CHF of zircaloy-4 significantly increased as contact angle decreased. This trend could be explained by existing CHF correlations incorporating wettability effect on pool boiling CHF, e.g., Kandlikar (2001). However, the CHF enhancements on the treated surfaces with small contact angles below  $10^\circ$  could not be explained with only the wettability effect.
- Liquid spreading phenomenon was observed on the surfaces with contact angles less than  $10^\circ$ . The experimental results above the CHF prediction with the wettability effect could be interpreted with liquid spreadability on the surfaces: (a) in the non-spreading regime, the CHF increased as the contact angle decreased; (b) in the small spreading regime, the CHF enhancement was greater than could be predicted by the wettability effect; (c) in the large spreading regime, the combination of different micro/nano multiscale structures on the zircaloy-4 surface contributed to greater CHF enhancement than non-combined multiscale structures. These results showed that the surface multiscale structures in the low-contact angle regimes of less than  $10^\circ$ , which had large effects on the liquid spreading ability, were the main influence on the CHF enhancement. In addition, these results suggested that the micro/nano multiscale structures of the surface promoting liquid spreading would be another main factor for the significant CHF enhancements.

## Acknowledgments

This work was supported by Nuclear Research & Development Program of National Research Foundation of Korea (NRF) funded by Ministry of Education, Science & Technology (MEST) (Grant code: MZ0706000050-08M0600-05010).

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